

The edifying impact of blending Game-Based Learning with Educational Robotics: A systematic review of empirical evidence

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Abstract. Game-based learning is an effective pedagogical approach with a demonstrated capacity to activate learner engagement, inspire motivation, and enhance the overall learning experience. The application of educational robotics has also attracted a lot of attention in recent years across educational levels and domains. Despite their appeal and the positive learning outcomes associated with such innovative pedagogies, the synergistic edifying impact of blending them remains largely unexplored. The aim of this study is to present a synthesis of empirical evidence on game-based learning and educational robotics. A systematic literature review is conducted focusing on empirical research published between 2019 and 2023. The analysis reveals prevalent methodological approaches and pedagogical theories framing learning and instruction, as well as the most widely employed robotics and gaming platforms. The study sheds light not only on the benefits of embracing game-based learning and educational robotics, but also on the barriers and challenges associated with adopting such innovative pedagogies. Ultimately, the study attempts to portray the impact of these approaches on learning and transferable skills development.

Keywords: Educational Robotics · Game-Based Learning · Pedagogical Frameworks · Skills · Competencies · Empirical evidence

1 Introduction

The integration of robotics in learning activities has gained substantial momentum in recent years, with several studies revealing that embracing educational robotics (ER) in the classroom can result in considerable instructional benefits and learning gains [9, 26, 34]. Game-Based Learning (GBL) constitutes another prevalent innovating pedagogy increasingly employed in educational interventions [24]. Through GBL activities, learners are presented with genuine opportunities for tinkering and inquiry-based learning commonly associated with positive emotional, behavioural, and cognitive learning outcomes, such as increased learner engagement, enhanced understanding and knowledge acquisition, and transferable skills development [24, 33].

The present study explores the impact of blending game-based learning and educational robotics (abbreviated to *GBL-ER*), placing an emphasis on the development of transferable skills, abilities, and competencies. Even though both pedagogical territories stand strong in the realm of educational technology, very few studies have attempted to explicitly explore the interaction between GBL and ER and the synergistic impact this combination can have on transferable skills development. As a result, these two innovative pedagogies have remained largely separate in the past [26]. This illuminates an opportunity for leveraging each other’s strengths to improve the learning outcomes and promote logical and critical thinking, problem solving, and social skills through playful and interactive activities [9, 40]. Furthermore, although the application of GBL-ER can serve as a versatile platform for enriching both practical skills (such as design, engineering, and programming) and soft skills (like computational thinking, critical thinking [7], problem-solving, teamwork and collaborative learning [1], communication, and leadership [18]), more emphasis is often placed on the former. This presents a gap in understanding the full potential of blending the use of GBL with ER toward transferable skills development. The focus on competencies and soft skills development is motivated by this gap, and the growing need for continuous upskilling and reskilling in light of the ongoing digital transformations across all levels of education and in workplace settings alike. Hence, a systematic and critical analysis of recent literature is deemed essential for educational researchers, instructors, and practitioners to make informed decisions about leveraging ER in GBL contexts, and vice versa.

After defining the research aim, objectives and research questions, the paper presents the educational profile of GBL-ER, and the methodology employed for conducting the systematic review, followed by a synthesis of key research findings. The paper concludes with a discussion of avenues of future research.

2 Research Aim, Objectives and Research Questions

Given the diversity in methodological approaches employed in empirical research, the educational theories and pedagogical frameworks applied, the participants’ characteristics and level of education, as well as differences in pedagogical design and implementation contexts, the present study aims to systematically review, evaluate, synthesise, and present empirical evidence drawing on recent literature on GBL-ER. To address the identified research gap, this study systematically reviewed empirical evidence in the field of GBL-ER across various educational levels and contexts. Two primary research objectives are being pursued:

1. To evaluate the most prominent research methods and methodologies; learning theories and pedagogical frameworks; robotics and games platforms; benefits (learning outcomes and positive impact) ascribed to GBL-ER as well as barriers (challenges or negative effects) that impede its effective implementation; and, ultimately, the impact of implementing GBL-ER on developing transferable skills, abilities, and competencies.

2. To identify avenues for future research towards establishing stronger theoretical foundations for GBL-ER, exploiting the constructive impact it can have on learners' knowledge, skills, behaviour and overall learning experience.

The following research questions were formulated based on the aforementioned research aim and objectives:

1. What methodological approaches are used in research exploring GBL-ER?
2. Which levels of education are empirically explored in GBL-ER research?
3. What are the underlying pedagogical theories framing GBL-ER activities?
4. What types of robotics platforms and games are used, and how are they combined to improve the educational experience?
5. What skills, abilities, and competencies are emphasised in empirical research employing GBL-ER?
6. What are the potential benefits (learning outcomes) and challenges (barriers) of GBL-ER?

The findings stemming from the analysis can serve as the cornerstone towards crafting a conceptual research framework for soft skills development through GBL-ER. This framework can guide future research endeavours leveraging and synthesising different strands of educational technology.

3 Educational Profile of GBL-ER

3.1 Game-Based Learning

Educational or serious games are widely employed in learning and skills development initiatives [10]. GBL involves learning through playing games [24], including massively multiplayer online games [5], mobile games [33], and immersive Virtual Reality (VR) games [43], amongst others. GBL approaches embrace a series of game mechanics including storytelling, narratives, role-playing, competitive tasks, points, badges, and reward systems to enrich the learning experience [32], make learning more fun, accessible, and 'seamless' [42, 48].

GBL is frequently associated with positive outcomes including: triggering creative thinking, problem-solving, analytical and communication skills [24]; prompting imagination and curiosity-driven learning [5]; promoting wellbeing, happiness, joyful and playful experiences [24]; activating learner engagement, motivation, participation and social interactions [57]; encouraging tinkering and learning through trial-and-error [24]; promoting goal orientation and enhancing learners' communication skills; activating interest through the use of multimedia and multimodal content; improving students' cognitive, meta-cognitive, and social skills, and the ability to retain newly learned information, hence offering an alternative to the conventional focus on structural learning and memorisation [24]. By blending entertainment and learning elements, serious games constitute a powerful educational medium, which can improve learners' emotional, behavioural and cognitive engagement, augment the level of understanding and knowledge acquisition, and enhance skills development [33, 42].

In addition to the vast array of benefits associated with GBL, there are still many barriers constraining their impact [33, 44]. Key challenges include the need for customization and personalization of existing games to fit individual learners' needs, (dis)abilities, and preferences, the challenge educators have for monitoring progress and providing timely and relevant feedback, and the scarcity and rigid nature of most games making them difficult to adjust to diverse settings [44]. Therefore, future research and development should emphasise the importance of prioritising human factors in GBL initiatives [42].

3.2 Educational Robotics

Physical and virtual robots, robotic simulations, and robots' digital twins, are becoming increasingly more accessible and affordable, making their use plausible beyond industrial settings, infiltrating our households, classrooms and workplaces, and creating intelligent educational and work spaces. Recent studies have showcased innovative educational applications combining robotics with Artificial Intelligence (AI) and VR, including physical robotic heads encapsulating affective robots responding to emotional cues and context-driven interactions [35] and digital twins of physical robots embedded in VR environments for educational purposes [43]. The use of social robotics is also emerging in educational milieus [1, 8, 21, 41, 53] providing novel opportunities to promote Education for Sustainable Development (ESD), including efforts to improve learners' attitudes toward recycling [11] and child rights education [15].

The influence of ER is observable in diverse settings, ranging from classrooms to ambient assisted living and learning spaces for individuals with special needs, and across a variety of domains, extending from software project management and planning skills [9] to Science, Technology, Engineering, Arts, and Mathematics (STEAM) education [3, 12, 16]. Educational research employing ER focuses on an array of aspects ranging from technical specifications, usability, effectiveness, and aesthetics, to programmable features, Human-Robot Interaction (HRI) design, technology acceptance, user engagement, and social dynamics of ER [?]. While predominantly targeting K-12 students in the age group of 6-18 years [26], educators are increasingly employing a range of robotics technologies, ranging from platforms such as LEGO, Arduino Learner, and REV, to programmable humanoid or social robots (e.g., Nao, Pepper, QTRobot, Cozmo, etc.), in an attempt to tailor learning across a broad range of learner groups, including pre-K-12 [53] and higher education [1, 7, 9, 34].

While recent research has illustrated the feasibility of ER, including social and humanoid robots [8], in diverse educational and training settings [7, 9, 40], it has also revealed that it is not always straightforward to obtain long-term interaction and engagement between robots and learners [53]. The key factors and conditions that may affect the impact of HRI on the overall learning experience include the length of deployment, the autonomy of robotic action, the duration of interaction time [53], the type of robot used [9], the task at hand [1], as well as the pedagogical models employed [34], among other factors. Therefore, further research is needed to continue exploring the realm of ER.

4 Methodology

4.1 Research Design

This research focuses on the edifying impact of game-based learning approaches and educational robotics on the learning experience. To address the research questions, a systematic literature review was performed using thematic and content analysis [57]. The empirical evidence was drawn from 56 journal articles and conference papers published in the 5-year period between January 2019 and December 2023. Employing a combination of content and thematic analysis was beneficial in the process of synthesising the extracted data into themes corresponding to each research question, while also capturing emerging patterns, prominent research clusters, as well as gaps in the literature. By using this approach, information from the studies included in the review was organised in thematic categories (e.g., level of education, pedagogical theory, gaming platform, robotics platform, etc.) which were subsequently analysed with the vision to inform future research, or identify specific areas where empirical evidence is limited and hence further case studies can be conducted.

4.2 Search strategy

The present study investigated peer-reviewed academic articles published in conference proceedings or academic journals from 2019 to 2023 in three major bibliographic databases: Springer, IEEE Xplore, and Elsevier Scopus. During the search process, papers from these three databases that matched a set of search terms related to *educational robotics*, *game-based learning*, and *skills*, were extracted. The following search strings¹ were used to obtain results from these three databases:

- **Springer and IEEE Xplore:** ("educational robot" AND "game-based learning") AND (skill* OR abilit* OR competenc*)
- **Elsevier Scopus** ((educational robot* AND game* based learning) AND (skill* OR abilit* OR competenc*))

4.3 Inclusion criteria

The purposeful combination of GBL-ER with skills development yielded a total of 122 articles as follows: Springer (n=24), IEEE Xplore (n=6), and Elsevier Scopus (n=92). A set of inclusion criteria was used to ensure that the selection process was objective and that the final corpus of papers could be identified for subsequent analysis.

- The studies had to be empirical research papers, published in academic journals or conference proceedings between January 2019 and December 2023, and written in English.

¹ A common query string was used but was adapted to each database's search syntax.

- Studies employing different methodologies and research methods (qualitative, qualitative and mixed methods) were included.
- Studies conducted in all educational levels across the world were considered, including primary, secondary and higher education, and vocational training.
- Articles exploring game-based learning and/or educational robotics were included even if they did not explicitly combine both approaches, and were tagged accordingly.
- Papers matching variations of the keywords specified in the query strings, including plural keywords and synonyms were included (e.g., skill, skills, skillset; robot, robots, robotics; competence, competency, and competencies).
- No restrictions were applied on the journals and conference proceedings, in an attempt to capture diverse viewpoints since the broader field of GBL-ER is defined in the intersection of multidisciplinary areas such as Human-Computer Interaction (HCI), Information Systems (IS), Technology-Enhanced Learning (TEL), and Humanities and Social Sciences (HSS), among others.

4.4 Exclusion criteria

The initial database search yielded a total of 122 publications which were considered for the second phase of the review. In this phase, the screening and selection steps were carried out based on a set of exclusion criteria. Initially, a table was created for each paper, including the paper’s title, abstract, and keywords, and a unique ID was assigned to each entry. The authors then independently read the abstracts and tagged each entry with the respective label ‘include’, ‘exclude’, or ‘indecisive’, along with a short statement justifying the reasoning for their decision. The authors then compared their classifications to discuss any mismatches, to reach consensus on entries labelled as ‘indecisive’, and to eliminate researcher bias. The following studies were excluded (some studies were flagged on more than one criterion):

- Book chapters, books, theoretical articles, systematic reviews, bibliographic studies, meta-analyses, editorials, etc. (n=19);
- Studies referring to the use of robotics in sectors other than education (e.g., industrial robots) or generic technology applications (e.g., wearable devices) without reference to GBL-ER (n=47);
- Studies focusing on teachers rather than students (n=6);
- Studies focusing on the technology (e.g., game mechanics or robot’s configuration) rather than learners/learning (n=9);
- Studies exploring gamification but not GBL (n=3).

Ultimately, out of the initial 122 papers, 66 were excluded following the collaborative screening phase, leaving 56 articles eligible for the third phase of the review. Fig. 1 presents a flow chart summarizing the filtering, eligibility screening, and final inclusion process.

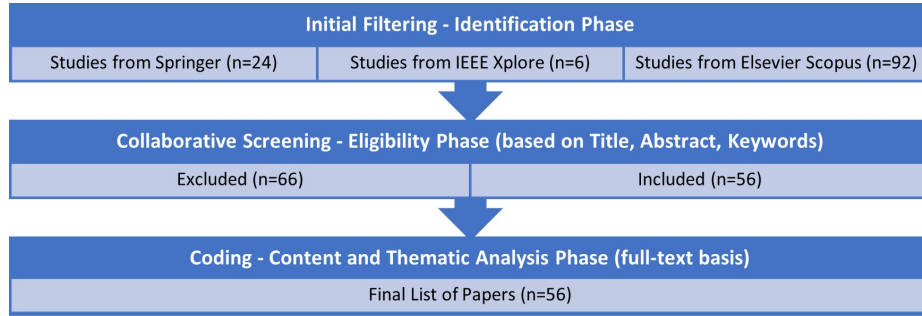


Fig. 1. The process of filtering, eligibility screening, and inclusion.

4.5 Coding and data analysis

At first, the authors independently analysed the final set of papers (n=56). During this process, key pieces of information relevant to the research questions were coded. The coding structure was recorded in a spreadsheet file, with rows representing the papers, and columns representing the codes (both deductively generated codes and emerging ones). These included the paper's ID, year of publication, type of study and research methodology, the level of education being studied, the pedagogical frameworks employed, relevance with robotics, humanoid robots, educational robotics, or GBL, combination of GBL-ER, reference to soft or hard skills addressed, and information on assessment strategies and intervention duration. Data analysis also included recording any surprising results or interesting observations in terms of student engagement and motivation, considerations regarding usability and accessibility, learning outcomes, and the potential benefits and challenges of each approach.

5 Results

This section presents our findings and attempts to provide insights into the research questions based on the results gathered by analysing the final set of 56 articles. To provide a high-level understanding of the research trends in this topic, we first analyse the chronological distribution of publications in the last 5 years. We then discuss findings related to the research methodologies used in studies addressing GBL and educational robotics. Furthermore, we present results on the level of education (where this was mentioned in the studies), the skills addressed, the pedagogical frameworks, assessment strategies, and the robotic platforms, games, and tools utilised in experimental evaluations. The impact of GBL and ER on the educational experience is further analysed by discussing profound benefits and barriers illuminated in the empirical studies under analysis.

5.1 Chronological and thematic distribution

The selected papers were classified based on the publication year and their relevance with ER, GBL, or both (Fig. 2). The yearly distribution of papers indicates an upward trend in the number of papers published per year, despite being limited by the number of databases and the range of years searched. More specifically, the number of papers tripled from 6 in 2019 to 18 in 2023. This indicates the increasing interest in exploring these approaches in educational contexts, as well as the widening utilisation of the underlying technologies.

The studies were also categorised into three thematic categories based on their relevance to ER, GBL, or a combination of both. Fig. 2, which presents the number of papers in each of these categories in a chronological distribution, suggests that a substantial number of studies involves a combination of GBL-ER. A total of 34 papers (61%) utilised an approach combining educational robotics with game-based learning, whereas 18 papers used ER only (32%), and 4 papers (7%) used GBL only. In 2023, the vast majority of the analysed papers (78%) combined both fields in their empirical research. This observation suggests an upward trend not just in the total number of papers per year, but also in the number of papers combining both fields as a medium to improve the educational experience.

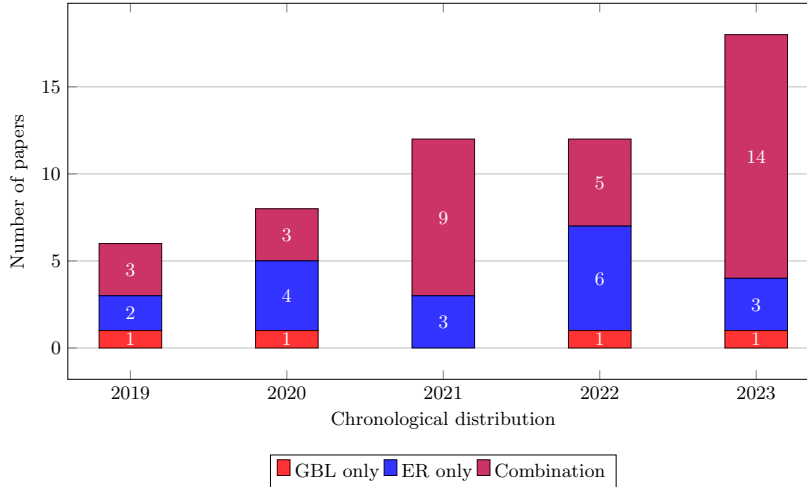


Fig. 2. Chronological and thematic distribution of selected papers.

5.2 Methodological approaches employed in empirical research

The first research question attempts to determine which methodological approaches have been used by the selected corpus of studies to explore GBL-ER. To this end, we have categorised the different methodologies used by researchers

in order to determine which are the most widely used ones and, equally, to identify gaps in the research methodologies utilised in recent studies. The results shown in Table 1 present the set of research methodologies as reported by authors (where applicable), sorted by frequency of use. It is worth noting that one publication may have reported none, one, or mixed methods, hence the numbers recorded in the table refer to frequency rather than number of unique papers.

The results show that the majority of studies have used a quasi-experimental approach (60%). Analysis of previous works and proof of concept approaches were employed as part of mixed-methods studies (10%). Less frequently used approaches included interviews and focus groups, and observations of students and other study participants in small-scale studies. Only one paper used expert evaluation. It is also noteworthy that all of these approaches used a limited time-frame in terms of data collection, which was often constrained to a few weeks. This may suggest a lack of studies attempting to reveal the long-term effects of ER and GBL on the learning experience, as well as the impact of each approach on diverse target audiences. In addition to short-term, most of the empirical studies appear to be small-scale with an average of 57 participants per study and only 4 studies engaging more than 100 participants (min=2, median=50, and max=232 participants). This presents a gap in conducting qualitative, ethnographic, and experiential studies focusing on a particular group of learners, and the need to launch longitudinal, mixed methods studies involving learners from diverse backgrounds.

Research methodology	Frequency
Quasi-experimental	34
Analysis of previous works	6
Concept paper	6
Interviews, focus groups	4
Observations	3
Expert evaluation	1

Table 1. Frequency distribution of research methodologies used.

5.3 Level of education

The second research question attempts to identify the levels of education in which ER and GBL are used. Fig. 3 displays a distribution of the different levels of education identified through the studied articles. Various case studies have utilised technologies, tools, and methodologies related to ER and GBL in diverse classroom contexts, in special education, and as part of specialised training. The majority of the studies (46%) reports findings from K-12 education, followed by studies conducted at undergraduate university level (13%). Fewer studies were conducted in special education contexts –mostly related to visual impairment–

(7%) and toward building specialised skills such as nursing, equipment training, or aiding refugees (7%). Surprisingly, none of selected studies were conducted in postgraduate education. This presents an empirical gap in GBL-ER research. Finally, a large set of studies did not explicitly identify the educational level or learners’ background. It can be assumed that these studies attempted to address multiple levels of education or engage a general audience.

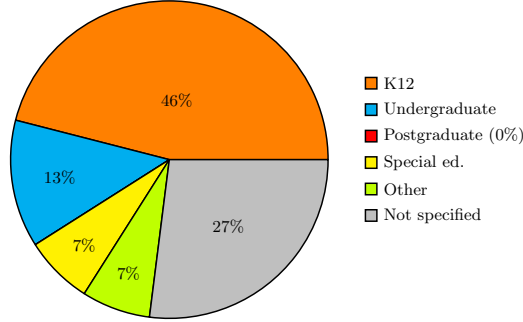


Fig. 3. Distribution of papers based on the level of education.

5.4 Pedagogical frameworks

The pedagogical frameworks and learning theories employed to leverage GBL or ER constitute the focus of the third research question. Theoretically framing the educational process is an important consideration and can inform curriculum design. It is noteworthy that the majority of the studies analysed did not explicitly mention any theoretical frameworks, neither provided any background information pertinent to the broader pedagogical approach. Only 24 (43%) papers discussed the pedagogical approach framing the GBL or ER learning activities employed in the studied context, including studies which have proposed and evaluated new pedagogical approaches to address unique contexts of use.

Among the pedagogical approaches utilised, the most prominent are *social constructivism*, *active*, *interactive*, *collaborative*, *situated*, and *project-based learning*, and *self-determination*. In addition, two of the studies mentioned the GAFCC (Goal, Access, Feedback, Collaboration, Challenges) model [31], which uses several theoretical constructs including *self-determination theory*, *goal setting theory*, *social comparison theory*, and *behavioral reinforcement theory*. A small subset of studies also utilised the SMART goal-setting model [20]. Despite the usefulness of these results, more research is needed to address the gaps which are evident in the pedagogical realm regarding the use of GBL and ER, across all levels of education. We argue that human-centred learning contexts employing GBL-ER can significantly benefit from existing learning theory and pedagogical frameworks, while novel theoretical explorations and pedagogical strategies can help illuminate the full educational potential of GBL and ER.

5.5 Robotics platforms and games

The studied articles feature a diverse set of robotics platforms and gameful activities which have been used either independently or combined in an educational context. Some studies have used integrated educational robotics platforms such as LEGO Mindstorms EV3 or LEGO Mindstorms NXT [58]. These sets feature multiple components required to build a robot using LEGO technology, such as building blocks, sensors, motors, and other parts. Such platforms often utilise both text-based and block-based programming tools to provide more flexibility to the students and to address wider audiences across multiple educational levels. These are diverse, high-quality educational robotics sets with a lot of support for their design and programming software, as well as third-party robot-building instruction sets. Conversely, other studies have reported the reduced availability of such sets –especially regarding spare parts– as well as their high cost, as two important issues when such technologies must be implemented at a larger scale (i.e., in school-wide curricula) [25]. Alternatives to these technologies include Arduino devices which are available at a significantly lower cost and higher customizability [38, 47].

The integration of various robotics platforms with other technologies was prevalent amongst the studied articles. Some studies utilised wearable devices like Myo armbands, while others employed mobile devices to provide a high-level GBL-ER platform [47]. The synergistic combination of multiple educational technologies seems to benefit learners. This illuminates avenues for future research leveraging the educational efficacy of mobile devices, AI, VR, and Augmented Reality (AR) [6] and combining them with robotics applications, or even capitalising on the immersive power of AR and VR in combination with Generative Pre-trained Transformers (GPTs) to enhance the user experience through simulated HRI or interactions with humanoid robots. Such technologies have leveraged commercial tools such as the Unity game engine, as well as commercially available platforms like the Metaverse. Given a particular problem context, any blend of technologies is plausible and may prove effective, as long as a strong emphasis is placed on human factors, skills development, and learner well-being.

The use of social robots has also been extensive, indicating the importance of HRI in providing a higher level of motivation and engagement, especially with younger audiences [11, 19, 21, 23, 28]. The combination of robotics and AI can potentially transform education, with the introduction of AI-assisted teaching [54] allowing a greater array of capabilities in addition to a higher level of HRI quality. Finally, only 12 out of the 56 studies (21%) considered the use of humanoid or social robots. While the analysed corpus of data is limited, we argue that this observation may instigate future research directions that take advantage of social and humanoid robots and their benefits to bring tangible results in terms of a more positive and engaging educational experiences [8, 9, 17].

From the perspective of GBL, several studies appear to utilise serious games in multiple contexts, including both general education and special education for visually impaired children, computing, mathematics, and languages, and training for skilled individuals such as nurses and athletes. It is evident that the use

of visual programming languages like Scratch and Blockly [45] can significantly improve the level of interaction with the underlying robotics technologies, but also with the game itself, however these must be adapted to specific learning needs. Examples of serious games utilised include the iDo serious game [39] as well as game-based competitive environments that involve maze-solving and line-following activities implemented by educational robots [38,49]. Finally, some studies have also blended ER and GBL to offer simulations/games for specialised skills training for dangerous work environments, managing cyberattacks or computer networks, creating an IoT system, or building electronic circuits [37,47,56].

5.6 Skills Development

All articles except one, identified or explored at least one skill, competency, or ability. To address our fifth research question, we categorised the identified skills as either *soft skills* or *hard skills* and, in turn, computed the frequency of each skill per category with the aim to identify the skills emphasised in GBL-ER research. We extracted a set of 17 soft skills and 16 hard skills promoted through either GBL, ER, or both. Tables 2 and 3 display the soft and hard skills, respectively, sorted by frequency.

Soft skills	Frequency
Computational thinking	14
Problem-solving	12
Teamwork, collaboration	8
Critical thinking	6
Creativity, originality	4
Green, environmental, sustainability skills	3
Engagement, attentiveness	3
Motivation	3
Exploration, experimentation	3
Navigation, spatial skills	3
Human rights education, equality	3
Self-regulated learning, self-efficacy	3
Decision-making	2
Empathy	2
Knowledge construction	1
Competitiveness	1
Flexibility	1

Table 2. Soft skills addressed in the reviewed publications, sorted by frequency.

The most widespread soft skills, promoted in the articles reviewed, are computational thinking, problem-solving, teamwork, collaboration, and critical thinking, while creativity, sustainability skills, engagement, motivation and exploration, self-regulated learning and self-efficacy were also highlighted. Soft skills

including decision making, empathy, flexibility, knowledge construction and competitiveness, were also captured, albeit to a lesser extent. Among the hard skills identified, the most popular was programming and computer science, followed by language skills, robotics, STEAM education, and design. Subject specific skills were also captured, such as mathematics and engineering, software testing and debugging, laboratory safety, nursing, computer networking, and others.

Hard skills	Frequency
Programming or computer science	14
Language, reading, speaking, vocabulary	11
Robotics	10
STE(A)M	6
Design	6
Mathematics	4
Engineering & mechanics	4
Software testing, debugging	2
Laboratory safety	2
Nursing	2
Object recognition	2
Computer networking	2
Computer-aided design	1
Computer-aided manufacturing	1
Data visualization	1
Cybersecurity	1

Table 3. Hard skills identified through the related works, sorted by frequency.

The analysis of the skills highlighted in the studied papers, reveals a pattern indicating that computing-related skills – both soft (computational thinking, problem solving) and hard skills (programming, robotics, design) – are among the most widely explored competencies. One plausible explanation for this observation is the close relationship between Computer Science and both educational robotics and Digital Game Based Learning (DGBL) [4,33]. Even though ER and GBL have been used in a wide variety of educational contexts and domains, their adoption still appears to be more prominent in the field of Computer Science due to their natural proximity. However, we argue that the benefits of ER and GBL may transfer to, and transform, the educational settings of other fields, especially when coupled with new, innovative pedagogical strategies.

5.7 Benefits and challenges

The final research question was formulated with the view to explore the potential benefits and challenges related to utilizing ER and GBL and their effectiveness as teaching and learning tools. GBL is characterised as an alternative

learning method [13] which can promote “*active education*” [50] and facilitate “*learning achievement*” and “*progressive skill improvement*”. It is more “*effective*” compared to other methodologies [45]. Participants reported that GBL activities “*improved their understanding*” [2, 50] and “*accelerated their learning proficiency*” [50] in various subjects. Furthermore, GBL was found to have a significant impact on children’s “*skills and motivation*”, while also “*minimizing costs*”, and providing positive outcomes in terms of “*enjoyment*”, “*task accomplishment*” and “*learning comprehension*” [27, 51].

ER has been used in conjunction with GBL to foster “*cooperation and critical thinking*” [29] and increase “*student engagement and concentration*” [36]. Learners perceived social robots as more trustworthy compared to other types of robots, and in some cases robots surpassed the popularity of human tutors [15, 28]. Coupled with AI, VR/AR, and mobile or wearable technologies, blending GBL-ER can provide a truly “*engaging educational medium*” [30, 52]. The findings stemming from our thematic and content analysis provide valuable directions for designing future learning spaces supported by these technologies, and guidelines for human-centred GBL-ER experiences in education [22], which can enable the desired transformation of the educational systems globally.

On the other hand, the current state of research in these domains reveals several shortcomings which must be addressed. A challenge regarding the adoption of these approaches – especially educational robotics – is the “*lack of qualified trainers, instructors, or teachers*” that can successfully deploy them [37, 51] and effectively take advantage [14] of these educational technologies. Blending GBL-ER approaches requires a “*relatively high level of technology literacy*”, which means that training must be provided to educators in order for these to be utilised effectively. Furthermore, several studies have cited the “*lack of resources available*”, especially in the case of ER [36, 51]. Given the variability of robot types and robotics platform to choose from, the diversity of game genres, types of games (digital, immersive, mobile, etc.) and game mechanics, the often complex production methods, the limited availability of user files, instructions and regulations, and other instrumental factors, blending GBL with ER may be characterised as a “*resource-demanding and high-maintenance approach*”. Lastly, it appears that the interaction design of these methodologies has not yet been perfected. The findings stemming from the analysis of the studied literature suggest that these barriers may be alleviated by actively involving learners in co-creating learning spaces. Substantial work is needed to understand how different social, contextual, pedagogical, technological, emotional, and cognitive factors can affect student engagement and motivation [15, 46, 55], and the learning outcomes.

6 Conclusions

This paper has explored an emerging field of research defined by blending two technology-enhanced pedagogical approaches: Game-Based Learning (GBL) and educational robotics (ER). The synthesised corpus of empirical studies was critically reviewed and analysed, focusing on research methods and methodological

approaches, theoretical models/constructs and pedagogical frameworks, robotics platforms and games platforms, benefits (learning outcomes and positive impact) and barriers (challenges or negative effects), and ultimately the impact of implementing the pedagogies on skills development. By reviewing and synthesising information from various related works, we have identified key methodological approaches, tools, pedagogical frameworks, and technological platforms used in these approaches. Our findings underscore the increasing interest in harnessing the combined benefits of GBL and ER, not only to enhance practical and subject-specific skills but also to foster soft and transferable skills, abilities and competencies such as problem solving, collaboration, communication, and creativity.

Our review has exposed empirical, methodological and theoretical gaps in studies addressing GBL-ER. Addressing these gaps is essential for understanding the full potential of integrating GBL and ER for transferable skills development. The findings revealed both the benefits as well as the challenges related to deploying these approaches. Despite the limited range of years studied and the study's scale and scope, our findings contribute to the broader conversation on innovation in educational methodologies and '*pedagogical engineering*'. It emphasises the role of human factors in technology-enhanced teaching and learning contexts, stresses the need for comprehensive skill development strategies through the combination of pedagogical frameworks and GBL-ER, and underlines the potency of the synergy between the two.

Looking ahead, it is crucial for researchers, practitioners, and educators to continue exploring these synergies. This may involve refining pedagogical models, crafting customised educational experiences, incorporating feedback from both learners and instructors, and also training them to blend various types of educational technologies. By doing so, we can better motivate and equip learners for the future workforce's demands and empower them with the varied skill sets needed for their success in an ever-changing digital world.

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